

A Packing of 73 Lines in $\mathbb{R}^{13}$

Limin Ge*

October 8, 2026

Abstract

We give 73 1-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 0.900257. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.899931, obtained by deleting 1 of the 74 lines of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 73 subspaces of dimension 1 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 1 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 0.900257 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 1$. The cell $N = 73$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 0.899931, obtained by deleting 1 of the 74 lines of the packing found by this search. The Rankin simplex bound for this cell is $1(13 - 1)/13 \cdot 73/72 = 0.935897$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n1N73.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 73 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:
900257961287484650375753366796225002055677818155
denominator:
999999999999894111811434789595709804477128690804

B Coordinates

73 blocks of 13 rows with 1 integers each; divide by 10^{12} . File: `grassmann/packings/m13n1N73.txt`.

-398368360416	332351874462	-26938593006	-354225649976
-496224452830	-402938820946	421443864350	-392009365939
-128715181000	-431409854967	-59733115534	58548025906
159043085538	28068427375	-19154707834	331168301031
-316637031634	22231218198	-760371853324	-118727222047
-229714949323	-235778378398	-195306173262	478133857719
-54932161942	-59778216874	226912331103	-28703449985
279172933122	381799426893	60345140188	-78451407799
-226186729921	-1641815918	218790824217	400926783219
11170955408	-230656056617	-157073725206	-29013032837
-346363080757	-161538986181	257101543916	415603289756
58519828208	622438902475	344456921191	-9817125004
-380186037573	-169073583147	93593847241	-260356077011
443822461050	-371319893210	63380716931	-160511264864
370956300796	-417788018501	45213442411	426778385295
-18447862519	-306120193929	236807865852	-397965899921
66114377286	-11411204930	496821835572	6488934442
-20965237752	49011439502	-362721473889	502577336193
491496287905	262962123381	189671399558	-212447362702
-36489994697	163912049241	-348367229486	315713553351
292047573620	-860493824	-365996545047	67035265932
-503892636553	17957526008	61521445570	-77434807668
-237037155987	-166290716276	-63940825715	390575577099
-63177209347	-559126551384	-247407226518	73236950094
15375005077	100155953769	-190999692050	197385363471
-133221661940	354244227529	346945562185	299351965328
101715777248	115373933129	-221365446706	2987444649
-263816644981	375313278161	3988755146	313006097647
-197108127527	235025574724	-235283719933	137969760157
-246132090302	15611703596	-1485239998	-5530100187
15378828973	427879943234	-32037164662	175157535654
45169626845	207152119033	-458003595349	-73333658663
202544764275	-218940319133	70373286037	71284234937
250687431374	-200754028123	430247870388	-538722314462
95555987568	670010855188	-169304767549	-377688042832
358241983245	108740880917	-353416930267	-509422363468
-605934823615	200895382138	-265481328117	-144426023580
419778251314	290210105787	220458586571	-348553129255
183405641289	-209249737169	3036688997	-72743780942
-152088345948	-241650644288	177903090689	239882992467
40114080199	180460578207	-500760537701	-359487652327
172112157686	-292418316939	209925437399	227732764364
239275499485	-250161799612	545706762262	-18484198488
343881337839	-278341773592	-243756194381	-21231158169
128448227800	190677982977	-254037105553	-149060384652
-205366646752	92318209394	-100253529031	-555083188318
55118159967	97412637961	76114034355	310213413213
475514610950	-214091938436	329224698400	-357430351917
346062543112	202273272107	239512221005	210348596449
571516515838	408954462171	-84767761840	-187663267771
71718901262	234637242528	116705237938	364649006095
-175412900264	-35470270154	65738236792	-300840712156
372131624098	-381853506208	-253025214080	-111430170487
302589511356	-11241190744	-422477123498	165730023795
-80111412781	401042168591	359189634237	273580737973
257761991183	-447490981533	-308489694225	-304232858785
-299287672058	-67018138057	51110577391	-433358203321
99412112411	-290418751276	-5284127509	84383611335
-351016101043	-272936215466	138275009393	-294120365660
-278817307432	139080961185	-131568690773	125562620734
-77062311576	237264722262	276503655289	-476300715042
276012732943	-280758282755	18350001380	-206697096943
-363434202656	-228006920458	-38627163890	-49598825829
31074653394	324850944072	253824504520	-29753261776
426051175561	-27187966118	246284578949	106031336650
-98962628374	49652745916	145205267491	72508618311
-92008695346	152967617390	-338025669125	20464096277
-32706960092	-18528132050	725983229470	233035093142
-555689353628	-646291385939	152763233579	-304213198471
82040998041	284282892637	-20472880225	109551004549

-469671564616	-345270214105	-299197141966	394165015432
364451837741	-443224642329	-55510735263	258623912792
-544304990667	135538397055	204429461208	-64911314377
201403962970	226354269706	142157936009	206414566895
345010375731	274140661712	-174718241883	-299578407229
-118588420893	216992232656	402803703684	-60484922022
-44491618657	-46161646213	-241402190275	486954410118
-321431978874	520719394487	-453731555233	154227748463
324056192255	-325601733138	-21277156089	-460765631632
31871461306	166925822188	-267409239510	35920400419
-621672827034	122363800786	352373515505	149993795728
-20811241185	211223589319	169309262365	75006895050
-188546489946	116250585129	139549094404	439944158134
49779774981	-93294373499	-89050291548	-179844516715
413881717340	97337205343	173389070395	366161457993
-18590686522	-574379459354	320332390223	259152562428
322012677447	-155242187162	211757733686	-344128138588
111376563004	201761882286	-484618845788	-171146172177
-275672203433	31869054250	-51101503831	94304086524
-278053231189	425097025543	-133020078272	282096843252
-215000040101	445416133219	-468401653118	303445181852
-673886961385	-263847534670	405808961132	-24551126397
-169910325868	138078573970	-54802145506	-471598338337
-235081460688	79068813498	103596586647	-103518608262
-119222056160	440002155756	57453806518	104311623014
330130752896	-190226067141	-376604457891	-3826982882
-120599765224	-235987627240	527981227468	532689968468
87039482787	-108011609779	109859985887	-78336986480
-295063257466	31840825662	458286060786	-21678155648
227521614069	227314472647	74644708671	-110952907013
-166900757286	594931157972	-442853623038	-180425113268
161864985358	-168610203427	-170815661509	433632624175
-330217090269	219001732368	-238413299315	3784512584
565207686622	156628407029	153567473654	321600187172
161395328173	-28064971524	-42583739158	-78675276951
-198122233378	295560185370	173695088579	-381721739347
-98808425268	-176833030429	-498483239867	78569286156
350059027426	521944634108	-127890918691	463900509641
-247863920193	-36174653257	-57574234722	-41525991916
223194573068	-90082552753	-14784213490	106861883480
-189414132565	-93494229530	50842502245	264241553949
178966022505	140188080360	115184837144	-300466762489
-67985016994	-121513015947	569054906518	124848065610
298287179988	-301976177601	308729255903	487615695255
318018159684	-480113178159	-382868940991	-331795163494
109258857266	479001175937	94533406674	157166013925
-51464190266	-66612735415	33234327030	212960358489
-396807452412	-51470423760	122261092012	-358797034861
-352443560975	374820407729	353635086107	143820372142
-478041560997	-201263179414	-173435713058	322710584782
-82203781872	-73337433273	-287037960484	-373755422674
-260126382090	-87975870510	410321291016	-331273303708
96752653861	651311877090	397301855821	16367118196
464771248982	-67078758734	255117226447	75519766561
320861611318	-51996256973	-152441852236	455023988828
-93972872854	160674252306	-351823049718	498453152541
-192444700507	176010017684	213806672160	235486865042
163104149859	172315861739	273036947582	467672859619
121431544353	-477361643958	-395430870053	-424222236826
266189241204	-239713346040	-77785806004	262481924774
170042738169	156227657582	-107496166807	9794916493
119579543735	-373511819730	-234478276248	5173901857
-330762408163	-37951458680	201672893280	-332572653775
371657011162	106110903538	291013617534	-594140752
-211560732242	-411434542072	-169929842584	54023670544
-412252537710	472340343005	-403507879707	415516284447
-225013260562	424851566630	-128333574851	246140724378
210945351867	128106934287	-276868470284	441005064341
351648383117	-215661816119	-77172555710	-68963186884
349003398433	-80114585552	129022721117	311629860060
-261442533982	366638361128	155517494911	492519540708
361258716543	-194599427948	26780231492	-62438384256
464677272056	-94164691814	640040085329	-192529223058
-4747388865	144178341308	84560169902	400251387504
21557856995	177922011845	-354276206885	-67004095055
151609731294	-133661898329	191064454227	-115755174242
-268913330248	176298632552	-419390277289	76526549160
9219572680	369069834687	505260753490	319680432633
236420199527	405451905668	209177402468	-6517612423
210398106785	-439578890498	100557071280	-17809444070
446488411400	340857029676	228368338400	17163700350
-169030888631	309250727559	-134894729187	384598232251
-473014304195	-112071028549	-574382440192	74326927343
-76598005502	-357131467308	184953380349	-545538327474
300640980269	-87758770292	-23086653307	-189145805311
55166366039	212490780312	156241389954	-316040992459
60943610114	239129803187	460077287	-217426311530
67355828961	374647250439	137966034684	157785950026
433334858516	393189881087	161223378768	446279756258
-194322945884	-156038605769	-32150916919	197746153671
392441731158	-548808934819	301105669812	2700687313
-508775717553	45347978294	247455299211	-524193276329
-275272981245	30902027283	402266607318	456556037290
448008483142	271373838458	220214488718	157408116760
245852811192	-37940699552	176353686443	168704574158
-131648993775	16885391565	428918631721	-97693460441
-241092350414	284286821780	-1306407078	391670132332
336337148602	-332522156707	-50914433803	130485528716
-136608698150	229235838068	286846032454	-29501457274
280790309832	-158075318759	-526647687138	518412022827
-8793831582	-499512854102	182913948193	-20831873259
92249785208	204971703259	367275510835	-6007280331

-116037743246	5096069426	7576581278	93639696633
-212689022942	103119196915	-515651588720	-460833624948
148720146433	-204228069163	315416984025	8683278802
240909713187	27421040751	-215104682447	-466330386346
-126472139037	43487331981	330953567593	13936804202
-334325648095	-128289451276	-206462140604	25375557759
-329038657745	19475568026	320390345312	-147044452391
-478705856626	281238762202	33348797290	-168479392083
-322669638085	749965939022	-62884694375	328385998370
-142451693538	-21228073192	344856672272	12646175509
428548139514	-192050736888	294455102297	510887189924
64836018810	310766182250	-295182893129	266526045203
-307482167600	-390764541347	-108394970911	61040928199
-51019117302	-198588578479	-135016677536	-537894744233
-87627056113	-236487327881	-8974521951	71198988788
239539162969	382717982281	136945903250	-333350165972
268126142979	210110312493	498504741342	-108943128191
-311831391513	-208634887139	-76966783052	-154618662905
-45135867508	-182195891731	-199093467730	-185050273282
187243655119	420293978546	693399816819	-398948336318
110566557149	-69699674503	-228502813927	46367386869
-286109259389	92150745700	145652130282	95940786973
437044825017	-504175107306	91008554003	-358715345385
627438484726	124584210183	305002775700	-387950131210
-170678987545	-243574862580	-39706394966	-517486355896
116701336862	343880433991	-651010151095	-50376538346
-84813845385	292442604784	-24732559709	665034091244
-83135316992	57727161708	-42390274687	28324113504
36864091621	70972432010	-61656421922	-121171708613
-332414688291	629659829535	88036270281	409407849360
-571794015631	499977794307	-213319108308	-20920686284
203647445545	-200246582341	-182251084810	13810739667
-312574360869	162584113765	-329745656352	-85177250985
153515099194	-110391090020	-187156971623	59947346742
-262753417664	321160193659	-441011383047	43082870729
98714963037	91203747947	-252600777961	-280511312403
122514542763	-114830893973	-285382880008	-110815691979
-223450952894	98024234165	-1799451592	-338994086510
-298366517672	216062854291	215056292042	-112765408259
-398488412450	2733235995	-491564064827	-517751761810
-426006997550	-285537025544	419556375933	54845330047
251020208282	21349576170	-267392970949	2008731429
-130496836736	176315596585	68529906700	-153791893918
339097087444	-413457724980	231512756269	-381726680899
-691181171837	259114849132	-167750411430	116829637936
-792411954	-516865583918	366394072804	-226531545231
38982705018	-65518380805	-290704095778	55112710771
-106594317592	-23676266006	-290441638911	-177513124627
-317776601108	-257690044580	-183764561797	193305229257
-75722843494	-111010593744	29109279836	543175751606
34683961963	-531411032809	201066419665	382882538587
-25504530706	125307201119	4919440500	-198011789669
155907723886	107730162766	-171240373307	64328719149
-461075439847	-333050311275	104267007090	-533726914542
100143640572	-79195712264	-35548232041	-323746562184
8768857106	121817486347	422682916522	-135086380466
335665786873	328125449399	775492079377	348939639525
-151146437138	-127285684427	47649536134	39866853173
351081879635	291272080980	7075937868	279562646724
-319884591160	-107191770628	-175170411251	-276827205070
-348490370443	-73515631531	310012356013	91151374531
478040747151	-47849956560	80339641732	249715080273
-108763743251	694430632874	176513116677	230925345839
-162370448506	174790196420	108370731969	
165430721150	340122186069	-296118109131	
-25939568969	-188816800967	231926871510	